

On-line determination of the oxygen demand of WWTP

The chemical oxygen demand COD is an important parameter for the monitoring and control of an industrial WWTP. However there was a big gap between on-line measurement and laboratory analysis until now. By determining the oxygen demand based on the oxidation of a sample it is now possible to reach a very good accordance between the international laboratory norm (ISO 6060) and the on-line measurement value – at a very low operational cost.

For the on-line measurement of COD there are quite a few different solutions. However they all have one thing in common: They 'try' to oxidize the sample by means of oxidants and to correlate the result to the standardized COD value. Every oxidant used has only a certain oxidizing power or oxidizing potential such as for example: ozone 2.05 V, H_2O_2 1.83 V or OH radicals 2.85 V.

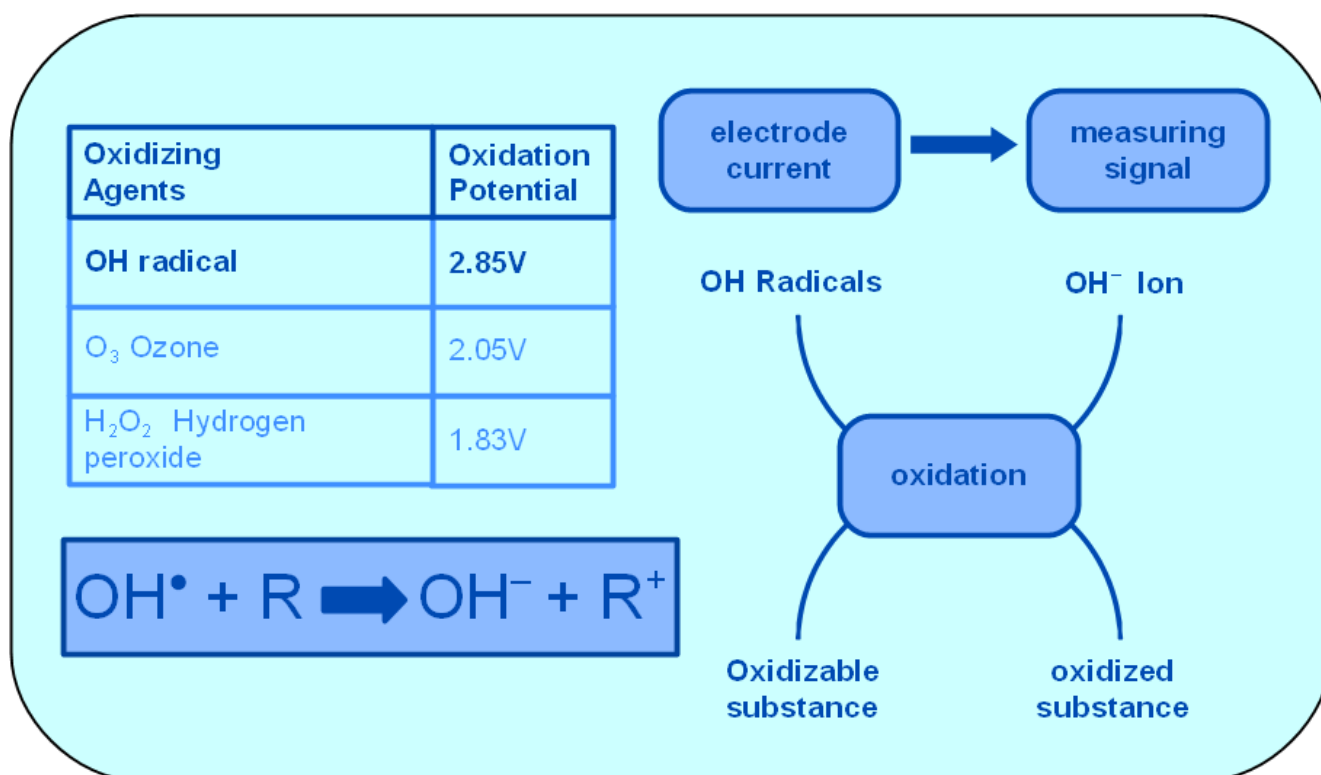


Fig.1 Comparison of oxidising potentials

The on-line analyser Elox100 uses OH radicals, which have a very high oxidizing potential to oxidize organic compounds in water. However it is important to point out that not all of the ingredients can be oxidized equally. It is necessary to differ between substances that are easy and hard to oxidize. In the chemical and petrochemical industry, with junction-measurements in the feed of the collection tanks or in the direct feed of

industrial WWTP many different matrix effects may occur, such as differently composed samples. These matrix effects complicate the complete oxidation of the homogenized sample with conventional COD analysers, or even make it impossible. So in any case, with the conventional on-line COD analysers, a correlation between the on-line result and a laboratory value has to be done. In the chemical industry the quality of the correlation is very poor – especially on a changing sample matrix. Here the operator has to decide between the quickly available (within 2 to 4 minutes) measurement value and the accuracy or incorrectness of the result. This is all the more problematic, as in general, control processes can only be achieved by on-line analysers. The laboratory determination of COD is quite expensive and time consuming (sample taking, sample preparation, analysis time and especially the provision of workforce itself).

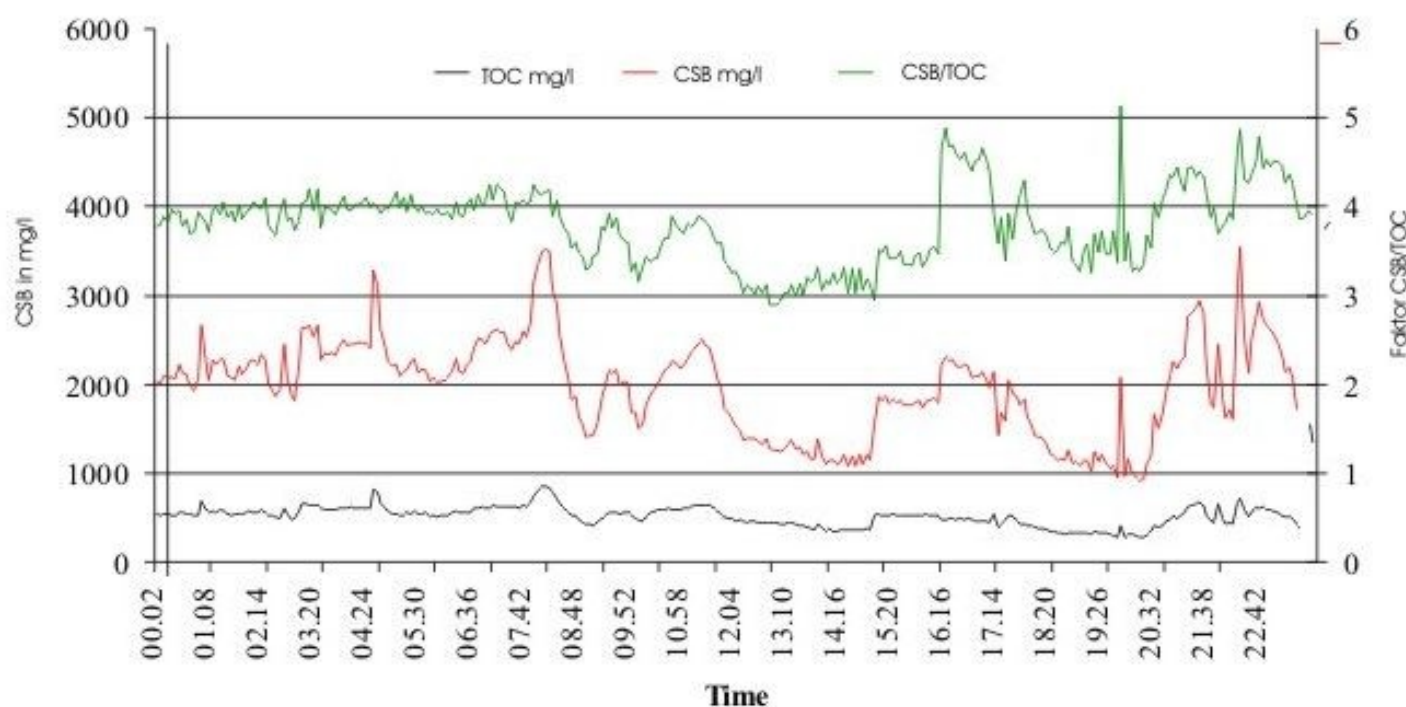


Fig.2 Measurements over 24 hour period of TOC, COD, TOC & COD

Additionally operating personnel have to observe safety regulations when operating with the aggressive chemicals used in the laboratory analysis of COD. The purchase, storage and disposal of aggressive chemicals are a further expensive criteria. Therefore in the chemical and petrochemical industry it has become more common within the last few years to measure the TOC (Total Organic Carbon) instead of the COD. With TOC analysers the method of thermal oxidation prevails. The other methods of oxidation, e. g. UV-digestion, have significant disadvantages due to their restricted handling of particles and/ or little oxidizing potential.

The thermal oxidizing methods separate into the thermal catalyst oxidation (usage of catalysts within the combustion reactor) and the thermal oxidation without catalysts. Since 1999 LAR AG offers a ceramic reactor enabling an oxidation at 1,200°C digestion temperature, which is completely catalyst-free.

Another difference to the other TOC analysers on the market is more obvious, no filtration is required before the original sample is to be oxidised, however the sample can be homogenized beforehand if required. Usually several different filtrations can be used, which filter the carbon or nitrogen in particles, enabling the

measurement of the diluted parts and leading to less findings.

The on-line analyser QuickCODo has the approved design of the QuickTOC analyser. However the detection is effected by an O₂-detector instead of a CO₂-sensor (IR-detector). Thus an exact determination of the oxygen needed to digest the sample is possible, and not only the amount of CO₂ produced when oxidizing it. This enables an almost correlation-free determination of COD without the matrix effects previously mentioned. At industrial WWTPs the COD is much more suitable as a performance indicator than the TOC, which only refers to the carbon in the compounds. This analyser has the following features:

- measurement without any chemicals or reagents
- single measurements within 2 to 4 minutes
- high salt concentrations possible
- no sample filtration necessary
- very low maintenance and operating costs
- also suitable for Ex Zone I and II

A combined or simultaneous measurements of TOC and TN_b is also possible. The measuring range can be between 20 to 150,000 mg/l COD – depending on the application. This on-line analyser is already used in several applications, such as the assessment of load in the feed of industrial WWTP in chemical parks, or in operations control at pharmaceutical plants for controlling the dosage of carbon at chemical WWTPs, or for the controlling of loads of anaerob reactors in chemical parks. In some of these applications, substances are detected which would be not oxidized to 100 per cent with the standardized COD, but show the real situation within the WWTP.

Conclusion: The on-line measurement of COD is suitable especially for applications, where no constant factor between the standard COD and TOC can be found and varying sample compositions go together with matrix effects. Furthermore, this measuring method makes the determination of the real pollution of industrial waste water, as well as, the fast and effective realization of control processes possible.